

BMJ Open Burden of disease using disability adjusted life years in the Middle East and North Africa (MENA) region: protocol of a systematic review

Simy Mathew,¹ Moustafa Sherif ,¹ Rami H Al-Rifai ,¹ Michal Grivna ,¹ Linda Östlundh ,² Brecht Devleeschauwer,³ Balázs Ádám ¹

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¹Institute of Public Health, College of Medicine and Health Sciences, United Arab Emirates University, Al Ain, UAE

²Örebro University Library, Örebro, Sweden

³Epidemiology and Public Health, Sciensano, Brussel, Belgium

Correspondence to

Dr Balázs Ádám;
badam@uaeu.ac.ae

ABSTRACT

Introduction In the Middle East and North Africa (MENA) region, changing demographic and epidemiological profiles have resulted in a diverse and shifting burden of disease (BoD). Disability-adjusted life years (DALYs), which combine years of life lost (YLL) due to premature mortality and years lived with disability (YLD), offer a valuable metric for assessing disease burden at the national level. While global burden of disease (GBD) estimates provide broad insights, national burden of disease (NBD) estimates offer country-specific data that can better inform tailored health policies and resource allocation. This systematic review protocol outlines our methodology for collating and analysing the NBD estimates in the MENA region using DALYs as the primary outcome measure.

Methods and analysis This review will adhere to the Preferred Reporting Items for Systematic Reviews and Meta-Analyses (PRISMA) guidelines. We will systematically search PubMed, Scopus, Web of Science and EMBASE for studies published from 1993 to 2024 that report national-level DALY estimates for diseases, injuries or risk factors in MENA countries. Eligible studies must report DALY estimates using national methodologies, while studies using exclusively GBD estimates will be excluded. Two independent reviewers will conduct title/abstract and full-text screening, data extraction and quality assessment using Standardised Reporting of Burden of Disease Studies (STROBOD), with disagreements resolved by a third reviewer when necessary.

Ethics and dissemination Ethical approval is not required for this review as it involves analysis of previously published data. The findings will be disseminated through publication in a peer-reviewed journal and presented at relevant academic and policy forums.

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BACKGROUND

Changes in health status and disease patterns in populations all over the world have been, and continue to be, key factors for concern over many decades. As an initiation to a global estimation of disease burden, the World Bank, under the collaborative efforts of the Harvard School of Public Health and

STRENGTHS AND LIMITATIONS OF THIS STUDY

- ⇒ A comprehensive search strategy has been developed, and the study selection and data extraction process were piloted with a research team composed of content experts.
- ⇒ This systematic review protocol follows the Preferred Reporting Items for Systematic Review and Meta-Analysis Protocols (PRISMA-P) guidelines.
- ⇒ Only articles published in English will be included in the review, thus relevant literature from local language journals may be excluded, resulting in language bias.
- ⇒ Variability in national data quality and methodological differences across studies may impact comparability.

the WHO, commissioned the first Global Burden of Disease (GBD) study in 1991 and the first report being published in 1993.¹ Estimates were updated in 1998, 2000, 2001, 2002 and 2008.² The Institute of Health Metrics and Evaluation, formed in 2007, updated the methods and techniques in 2010, including applying them to previous time series as well.³ Currently, the newest estimates are published in GBD 2021.²

Burden of Disease (BoD) refers to the measures of mortality and morbidity, pertaining to diseases, injuries and risk factors. The concept of disability-adjusted life years (DALYs) was first introduced in the 1990s by Murray as a comprehensive single measure to quantify the effect of diseases, injuries and risk factors.⁴ The calculation of DALYs is the summation of the years of life lost (YLL) and the years lived with disability (YLD). This enables the use of a common metric of years that permits comparison across populations on national, regional and global scales.^{5,6} DALYs are implicit in health programme planning on regional levels as

they allow assessment of the health gap and allow for assessment across multiple time periods.⁷

GBD data has been used extensively to describe global trends and project health issues. GBD methods are as inclusive as possible and use all publicly available data from each country and region.^{8–9} NBD studies have been taken up by many countries in Europe and in Australia.^{10–13} Research has shown that the health burden as reported by the GBD is not always prioritised in health planning and resource allocation in many countries.¹⁴ In such instances, a NBD estimation would facilitate effective health planning as it would help to narrow down the health issues that need to be prioritised. NBD studies would also rely on local data sources, which may not be publicly accessible and would also allow for identification of gaps in data and assessment of quality of data.¹⁵ Well-planned NBD studies would thus contribute to better estimates in the GBD.

BoD studies at national levels for various diseases and risk factors have been estimated using the GBD data for the Middle East and North Africa (MENA) region. These estimates are predicted based on varying levels of evidence from the country itself and on models generated by the GBD data.^{16–18}

RATIONALE

The majority of available literature use data obtained from the GBD, injuries and risk factors data over previous years. The determination of health priorities at a national or regional level requires data from the region rather than projections based on global databases. To estimate the BoD in the MENA region, it was proposed to conduct a formal systematic review to assess the available literature, and to identify and evaluate the BoD studies with DALY calculations at national/regional/local levels among the countries in the MENA region.

AIM

The aim is to conduct a systematic review of published scientific literature for BoD studies using local or national data sources, conducted in the MENA Region.

OBJECTIVES

The specific objectives of this review are:

Primary objectives

- To identify the BoD studies performed calculating DALYs) and/or its components of YLL and YLD in the MENA region.

Secondary objectives

- To identify the countries that have performed these studies.
- To identify the diseases, injuries and risk factors for which BoD calculations have been performed and the DALYs associated with them

- To identify the methodologies used to calculate DALY and/or its components.
- To identify the data sources used for acquiring required data within the identified studies and to assess the quality of these data sources.

RESEARCH QUESTIONS

Which countries in the MENA region have BoD calculations using DALYs or its components? What are the diseases/risk factors included in these BoD calculations? What methodologies were used to calculate DALYs? What data sources were required in the BoD studies?

MATERIALS AND METHODS

Registration of protocol

This protocol adheres to the Preferred Reporting Items for Systematic Review and Meta-Analysis Protocols (PRISMA- P) 2015 statement¹⁹ (online supplemental file 1). The protocol has been registered in the International Prospective Register of Systematic Reviews (ID: CRD42024498688). On completion of the systematic review, it will be reported adhering to the PRISMA 2020^{20–21} statement guidelines.

Eligibility criteria

To review the information on BoD studies, the population, exposure, comparison and outcome (PECO) framework was used to develop the criteria for selection and inclusion of studies.

- Population
Total population or a subgroup of the population of any country or group of countries in the MENA region. The MENA region typically consists of the following countries: Algeria, Bahrain, Djibouti, Egypt, Iraq, Israel, Jordan, Kuwait, Lebanon, Libya, Morocco, Oman, Qatar, Palestine, Saudi Arabia, Sudan, Syria, Tunisia, United Arab Emirates, West Bank & Gaza and Yemen.²²
- Exposure
Any disease (communicable/non-communicable), injury or risk factor to health conditions.
- Comparison
There will be no comparator to the outcome. These studies focus on individual diseases or disease groups or risk factors in their assessment; hence no comparisons are considered.
- Outcome
DALYs and/or its components of YLLs and YLDs attributable to any disease or risk factor.

Types of studies

Inclusion criteria

All studies that include direct calculation of YLLs, YLDs or DALYs for any disease, injury or risk factor using national, regional or local databases of mortality and morbidity. Letters, commentaries, opinion articles, published abstracts and conference proceedings will not be used.

Only articles in the English language will be considered for the review.

Exclusion criteria

Studies that calculate the BoD with outcomes other than DALYs, YLLs or YLDs.

All studies report DALY estimates from other sources than national, regional or local databases, such as the GBD data, Global Cancer Observatory or other global databases.

Publication period

DALYs were first described in the 1990s by Murray for the GBD study, which was a collaborative project of the World Bank and the WHO.¹⁴ Therefore, all articles from January 1990 till December 2024 will be included in this systematic review.

Data sources and search strategy

Four major electronic databases: PubMed (NLM), Scopus (Elsevier), Web of Science (Core Collection) and EMBASE (Elsevier) will be systematically searched to identify eligible studies published in peer-reviewed scientific journals (LO, SM and MS). Preliminary search terms were identified and tested according to the PECO statement and the defined inclusion and exclusion criteria. Subject matter experts (BÁ, RHA-R and MG) reviewed the final search term selection. PubMed's MeSH database will be included to systematically incorporate the search terms and their variations (online supplemental file 2). A comprehensive literature search will be conducted using the search terms, MeSH terms and their combinations, applying the inclusion and exclusion criteria and using the various Boolean operators, to retrieve potential articles. The search string will be generated in PubMed to include the terms 'DALYs', 'YLL' and 'YLD' and will be converted to strings used for the other databases.

In addition, the reference lists of all included studies and identified reviews, if any, will be manually searched.

Study identification and screening

The retrieved search citations will be imported into Covidence software for the automatic removal of duplicates and the screening process.²³ Any further duplications will be identified and removed manually during screening. A PRISMA flowchart will document the screening and selection procedure.

The identification of eligible studies will be a two-step procedure. The titles and abstracts of all the retrieved literature will be screened manually by two independent reviewers (SM, MS) and categorised as relevant, not relevant and maybe relevant. The articles that fall into the 'maybe relevant' category will be resolved through discussion between the two reviewers (SM, MS). In case of discrepancies, a third reviewer (BÁ) will be requested to resolve the articles in conflict.

On completion of the title and abstract screening, full text articles will be uploaded into Covidence, thoroughly screened and independently assessed by two reviewers

(SM, MS). Any discrepancies will be resolved by a third reviewer (BÁ/RHA-R).

Data extraction

A data extraction sheet will be created in Microsoft Excel for Microsoft 365 with all the relevant details to be extracted. The sheet will then be piloted on five eligible articles to assess the completeness of the information to be retrieved. Data extraction will be completed by a single reviewer (SM), with 20% of the papers reviewed by an independent reviewer (MS) for validation. Any disagreements will be resolved by a third reviewer (BÁ/RHA-R/MG), and conclusions will be applied throughout the data extraction process.

The variables to be extracted are proposed as follows: authors' names, title of the article, year of publication, country and/or region, disease group (communicable/non-communicable/ injury), specific disease, specific risk factor, reference population, reference year, sources of required data, method used for DALY, YLL, YLD calculation, life tables used for calculation of life expectancy and source of disability weights, estimated DALY, YLL, YLD.

Quality assessment

Two independent reviewers (SM, MS) will do a quality assessment for each study, with a third reviewer being consulted for discrepancies. This will be assessed using the Standardised Reporting of Burden of Disease (STROBOD) checklist, which is a reporting guideline.^{24–26} STROBOD will be used as there are no risk of bias assessment tools currently available to assess the quality of BoD studies. Any discrepancy will be resolved in consultation with a third reviewer (BD).

Data synthesis

Based on the extracted data, a summary table and narrative description of the findings will be provided. The narrative synthesis will adhere to the synthesis without meta-analysis guidelines.²⁷ The summary would focus on the various geographic locations, diseases and/or risk factors and data sources and methods used. DALY and/or YLD and/or YLL will be reported for each disease/risk factor by region or country.

Considering the possibility of a wide variety of diseases for which the BoD calculations was performed across the MENA region, only limited scope for a meta-analysis is anticipated. However, if comparable data on certain diseases or disease groups are identified, meta-analysis will be performed in additional studies.

Ethics and dissemination

This systematic review will include data from published literature and will not involve the use of any data that could identify with individual participants; hence ethical approval is not required. The results of the proposed review will be presented at conferences, published in a peer-reviewed journal and will be made publicly available in electronic format.

X Michal Grivna @dr_michalgrivna

Collaborators Not applicable.

Contributors BĀ, SM, MS, RHA-R, MG, BD and LÖ were involved in conceptualisation and protocol development. BĀ, SM, MS, RHA-R, MG and LÖ have developed the search strategy. Literature search will be conducted by SM, MS and LÖ. Screening will be performed by SM and MS, with BĀ resolving conflicts. SM and MS will extract data, which will be validated by BĀ, RHA-R and MG. Quality of the studies will be assessed by SM and MS with BĀ resolving conflicts. BĀ is the guarantor.

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ORCID iDs

Moustafa Sherif <http://orcid.org/0000-0002-8781-5098>

Rami H Al-Rifai <http://orcid.org/0000-0001-6102-0353>

Michal Grivna <http://orcid.org/0000-0003-2349-0922>

Linda Östlundh <http://orcid.org/0000-0001-5091-604X>

Balázs Ádám <http://orcid.org/0000-0002-9765-7142>

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